# Mikhail Razakov

## Software Engineer — Systems & Architecture

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I build systems where state consistency and transformation correctness matter. I work on symbolic computation in SharpChecker at ISP RAS; independently built UniversalToolchain and the offline-sync botay! app, and previously worked with LLVM/C++ at MCST.

2022: Compilers / VMs · 2023: CIL + native x86-64 · 2024–2025: AST / IR / backends → toolchain · 2026 →: LLVM + program analysis

**Compiler infrastructure**
UniversalToolchain: planned composition, a separate verifier, and interpreter/CIL backends.

**End-to-end product**
botay!: browser/Android clients, offline operation replay and validated planning witnesses.

**Reliability**
VpnMediator: durable payment evidence, idempotent access operations and quarantine.

## EXPERIENCE

**October 2026 — present**

### ISP RAS — SharpChecker development

- Work on symbolic computation in SharpChecker, a C#/.NET static-analysis tool.

**July — August 2026**

### MCST — Compiler Engineering Intern

- Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety.
- Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.

## PROJECTS

### UniversalToolchain / Wist2

Built one semantic planning boundary: feature dependencies, conflicts and typed backend routes become an immutable LanguagePlan before runtime execution.

A separate verifier checks exact package identity and routes; architecture gates and a 1,324-test manifest passed at the inspected 9 Oct 2026 revision.

### botay! / student-execution-os

Built web/Android task planning with separate owners for external evidence, canonical user state, derived schedules and LLM suggestions.

Offline commands replay idempotently; feasibility requires independently checked witnesses. Signed Android release 0.8.3 passed its workflow on 10 Oct 2026.

### VpnMediator

Persisted payment-provider evidence before subscription/access changes; idempotent entitlements and leased reconciliation handle retries and uncertain outcomes.

Ambiguous remote/local divergence is quarantined for review; recovery includes backup/restore and rollback safeguards. Deployment scale is not asserted.

## SKILLS

**Languages / platforms**
C++23, C#/.NET, Python, Rust, Java, JavaScript, SQL, Linux

**Systems design**
dependency planning, module contracts, state consistency and recovery

**Product / backend**
offline-first sync, Android, SQLite/PostgreSQL, payment workflows, migrations

**Testing**
LLVM Verifier, differential execution, invariant checks, idempotency and regressions

## EDUCATION

HSE University — Software Engineering

## RECOGNITION

**MCST LLVM/C++ selection**
Organizer correspondence: only 5.0/5.0 score and only solution passing all 104/104 tests among 49 submissions.

**LangDev'26 — Build the Language, Then Make the Abstractions Disappear**
Delivered the UniversalToolchain architecture talk at LangDev on 9 October 2026.

Moscow / remote